

FUNCTIONAL REQUIREMENTS DOCUMENT

OTT Streaming Intelligence — Power BI

Project Name	OTT Streaming Intelligence
Tool	Microsoft Power BI
Document Type	Functional Requirements Document
Version	1.0
Data Source	OTT Subscription & Customer Analytics
Data Model	Star Schema — 1 Fact + 6 Dimensions

1. Project Overview

The OTT Streaming Intelligence project is an interactive Power BI reporting solution designed to analyze OTT subscription and business performance. The dashboard provides a centralized view of revenue, customers, subscriptions, platforms, plans, content, and geographic performance.

2. Business Objective

- Monitor overall OTT revenue performance.
- Track customer activity and subscription behavior.
- Compare performance across OTT platforms and subscription plans.
- Analyze new subscriptions, renewals, and customer churn.
- Identify high-performing content genres and content types.
- Analyze revenue by region, country, and other geographic attributes.
- Provide time-based analysis using monthly, MoM, YoY, YTD, and MTD measures.
- Provide an interactive and easy-to-use reporting solution for business stakeholders.

3. Dataset Requirements

The dashboard will use the following cleaned Star Schema tables:

Table	Purpose
FactSubscriptionSales	Transaction, revenue, subscription and churn analysis
DimDate	Date and time-intelligence analysis
DimCustomer	Customer status, segment and customer attributes
DimPlatform	OTT platform performance
DimPlan	Subscription plan and billing analysis

Table	Purpose
DimContent	Genre, language, content type and title analysis
DimLocation	Country, city, state and regional analysis

4. Dashboard Requirements

The report will contain four dashboard pages. Each page will answer a specific set of business questions while maintaining a consistent visual design and navigation experience.

4.1 Executive Overview

Purpose: Provide a high-level view of overall business performance.

KPIs: Total Revenue • Total Customers • Active Customers • Churn Rate • Renewal Rate

Visuals:

- Monthly Revenue Trend — Line Chart
- Revenue by Region — Bar Chart
- Revenue by Platform — Column Chart
- Revenue by Plan — Column Chart

Filters: Year • Region

4.2 Sales & Platform Analysis

Purpose: Analyze revenue and transaction performance across platforms and plans.

KPIs: Total Revenue • Total Transactions • Average Transaction Value • Revenue per Customer

Visuals:

- Revenue by Platform — Column Chart
- Transactions by Platform — Bar Chart
- Revenue by Plan — Column Chart
- Revenue by Billing Cycle — Donut Chart
- Monthly Revenue Performance — Line Chart

Filters: Year • Platform • Plan • Billing Cycle • Region

4.3 Customer & Subscription Analysis

Purpose: Understand customer activity, subscription behavior, renewals and churn.

KPIs: Total Customers • Active Customers • Churned Customers • Churn Rate • New Subscriptions • Renewal Subscriptions • Renewal Rate

Visuals:

- Customer Status — Donut Chart
- New vs Renewal Subscriptions — Column Chart
- Customer Segment — Bar Chart
- Churn by Segment — Column Chart
- Subscription Trend — Line Chart

Filters: Year • Customer Segment • Customer Status • Subscription Type • Platform • Plan

4.4 Content & Geographic Analysis

Purpose: Analyze content performance and geographic revenue contribution.

KPIs: Total Revenue • Total Transactions • Revenue per Customer

Visuals:

- Revenue by Genre — Bar Chart
- Revenue by Content Type — Donut Chart
- Revenue by Language — Bar Chart
- Revenue by Region — Bar Chart
- Revenue by Country — Map / Bar Chart
- Top Content — Bar Chart

Filters: Year • Genre • Content Type • Language • Region • Country • Platform

5. Filters & Slicers

The report should provide relevant filters without overcrowding the dashboard.

Dashboard Page	Required Filters
Executive Overview	Year, Region
Sales & Platform Analysis	Year, Platform, Plan, Billing Cycle, Region
Customer & Subscription Analysis	Year, Customer Segment, Customer Status, Subscription Type, Platform, Plan
Content & Geographic Analysis	Year, Genre, Content Type, Language, Region, Country, Platform

6. Interactivity Requirements

- Slicers should dynamically filter the relevant visuals.
- Visuals should support cross-filtering and cross-highlighting.
- Date analysis should support Year → Quarter → Month drill-down where applicable.
- Geographic analysis should support Region → Country → State → City drill-down where applicable.
- Content analysis should support Genre → Title drill-down where applicable.
- Tooltips should provide additional context without overcrowding visuals.
- Page navigation should allow users to move between dashboard sections easily.
- A reset-filter option should return the page to its default state.

7. Data Model Requirements

The report must use a Star Schema with single-direction relationships from dimensions to the fact table.

Dimension	Relationship	Fact
DimDate[Date]	1 → *	FactSubscriptionSales[Date]
DimCustomer[Customer_ID]	1 → *	FactSubscriptionSales[Customer_ID]
DimPlatform[Platform_ID]	1 → *	FactSubscriptionSales[Platform_ID]
DimPlan[Plan_ID]	1 → *	FactSubscriptionSales[Plan_ID]
DimContent[Content_ID]	1 → *	FactSubscriptionSales[Content_ID]
DimLocation[Location_ID]	1 → *	FactSubscriptionSales[Location_ID]

8. KPI & Measure Requirements

The following measures are required for dashboard analysis. Measures should be created in the dedicated DAX_Measures table.

Measure	Business Definition
Total Revenue	Total net revenue generated
Total Transactions	Number of unique subscription transactions
Total Customers	Number of unique customers with transactions
Active Customers	Customers with Active status
Churned Customers	Customers with Churned status
Churn Rate	Churned Customers ÷ Total Customers
New Subscriptions	Transactions classified as New
Renewal Subscriptions	Transactions classified as Renewal
Renewal Rate	Renewal Subscriptions ÷ Total Transactions
Revenue per Customer	Total Revenue ÷ Total Customers
Average Transaction Value	Total Revenue ÷ Total Transactions
Revenue Contribution %	Revenue contribution within the selected platform context
Previous Year Revenue	Revenue for the comparable previous-year period
YoY Revenue Growth %	Year-over-year revenue change
Revenue YTD	Year-to-date revenue
Revenue MTD	Month-to-date revenue
Previous Month Revenue	Revenue for the previous month
MoM Revenue Growth %	Month-over-month revenue change
Active Customer %	Active Customers ÷ Total Customers
Total Platforms	Number of unique OTT platforms
Total Discount	Total discount recorded

9. Core DAX Measures

Total Revenue

```
Total Revenue =  
SUM(FactSubscriptionSales[Net_Revenue])
```

Total Transactions

```
Total Transactions =  
DISTINCTCOUNT(FactSubscriptionSales[Sale_ID])
```

Total Customers

```
Total Customers =  
DISTINCTCOUNT(FactSubscriptionSales[Customer_ID])
```

Churn Rate

```
Churn Rate =  
DIVIDE(
```

```

        [Churned Customers],
        [Total Customers],
        0
    )

```

Revenue per Customer

```

Revenue per Customer =
DIVIDE(
    [Total Revenue],
    [Total Customers],
    0
)

```

YoY Revenue Growth %

```

YoY Revenue Growth % =
IF(
    ISBLANK([Previous Year Revenue]),
    BLANK(),
    DIVIDE(
        [Total Revenue] - [Previous Year Revenue],
        [Previous Year Revenue]
    )
)

```

Revenue YTD

```

Revenue YTD =
TOTALYTD(
    [Total Revenue],
    DimDate[Date]
)

```

MoM Revenue Growth %

```

MoM Revenue Growth % =
DIVIDE(
    [Total Revenue] - [Previous Month Revenue],
    [Previous Month Revenue],
    0
)

```

10. Export Requirements

- Users should be able to export summarized visual data to Excel/CSV where permitted.
- Report pages should be exportable to PDF for sharing.
- Exports should respect the active slicer and filter selections.
- Exported output should retain clear titles and readable formatting.

11. Assumptions & Limitations

- Sale_ID represents a unique subscription transaction.
- Net_Revenue is the primary field used for revenue KPIs.
- DimDate is the official date table for time-intelligence calculations.
- Missing non-key attributes are retained when the correct value cannot be reliably inferred.
- Missing Net_Revenue values are not artificially reconstructed when required inputs are unavailable.
- 2026 contains partial-year data and should not be interpreted as a complete annual period.
- Churn analysis is based on the available customer-status information.

12. Acceptance Criteria

- All four dashboard pages are created according to the approved requirements.
- Required KPIs return accurate results and respond to filters.
- Star Schema relationships are active and correctly configured.
- Required slicers and interactions work as expected.
- Time-intelligence measures return valid results for supported periods.
- No unintended blank categories or broken relationships appear in key visuals.
- Dashboard formatting and navigation are consistent across pages.
- Results are validated against the underlying cleaned data.
- The final dashboard is understandable to business stakeholders without technical assistance.

Document Status: Ready for Dashboard Development